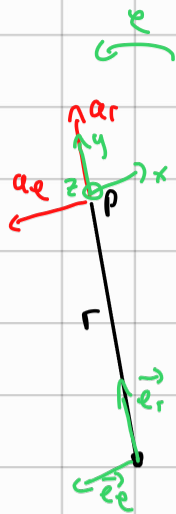


IMU pendulum readings



Dynamic forces:

$$a_r = \ddot{r} - r\dot{\theta}^2, \quad r = \text{const}$$

$$\rightarrow a_r = -r\dot{\theta}^2$$

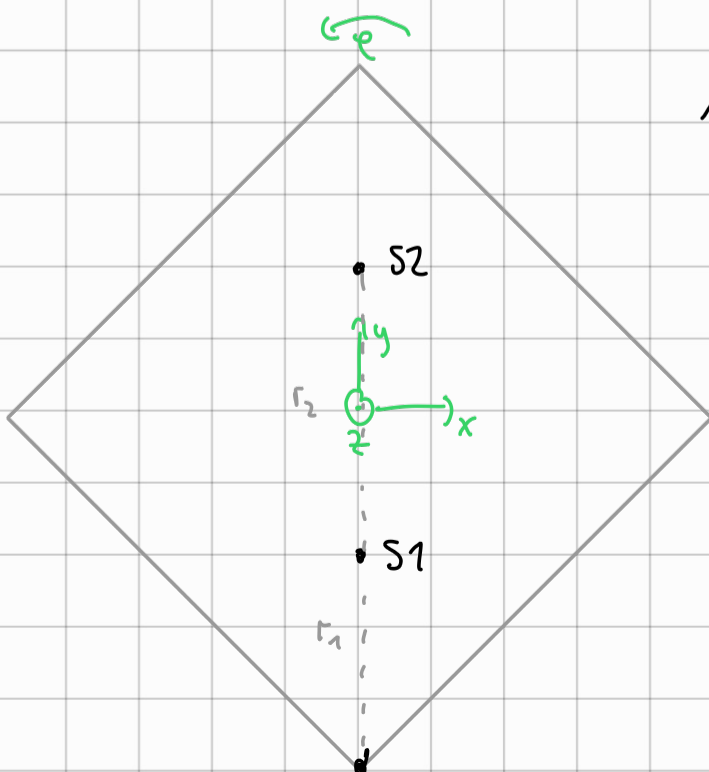
$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}, \quad r = \text{const}$$

$$a_\theta = r\ddot{\theta}$$

$$\Rightarrow \vec{a}_{xyz} = \begin{pmatrix} -a_\theta \\ a_r \\ 0 \end{pmatrix} = \begin{pmatrix} -r\ddot{\theta} \\ r\dot{\theta}^2 \\ 0 \end{pmatrix}$$

$$\Rightarrow \vec{g}_{xyz} = \begin{pmatrix} -\sin\theta \cdot g \\ -\cos\theta \cdot g \\ 0 \end{pmatrix}$$

$$\vec{f}_{xyz} = \vec{a}_{xyz} - \vec{g}_{xyz} = \begin{pmatrix} \sin\theta \cdot g - r\ddot{\theta} \\ \cos\theta \cdot g + r\dot{\theta}^2 \\ 0 \end{pmatrix}$$



Applied to Cube to figure out theta based on Accelerometer:

$$\vec{a}_i = \begin{pmatrix} \sin\theta \cdot g - r\ddot{\theta} \\ \cos\theta \cdot g + r\dot{\theta}^2 \\ 0 \end{pmatrix}$$

Subtract to remove dynamic forces:

$$\vec{a}_1 - \frac{r_1}{r_2} \vec{a}_2 = \begin{pmatrix} \sin \varphi \cdot g - r_1 \ddot{\varphi} - \frac{r_1}{r_2} \left(\sin \varphi \cdot g - r_2 \ddot{\varphi} \right) \\ \cos \varphi \cdot g + r_1 \ddot{\varphi}^2 - \frac{r_1}{r_2} \left(\cos \varphi \cdot g + r_2 \ddot{\varphi}^2 \right) \end{pmatrix}$$

$$\begin{pmatrix} \ddot{x}_1 - \frac{r_1}{r_2} \ddot{x}_2 \\ \ddot{y}_1 - \frac{r_1}{r_2} \ddot{y}_2 \\ \text{0} \end{pmatrix} = \begin{pmatrix} \sin \varphi \cdot g \cdot \left(1 - \frac{r_1}{r_2} \right) - \cancel{r_1 \ddot{\varphi}} + \cancel{r_1 \ddot{\varphi}} \\ \cos \varphi \cdot g \cdot \left(1 - \frac{r_1}{r_2} \right) + \cancel{r_1 \ddot{\varphi}^2} - \cancel{r_1 \ddot{\varphi}^2} \\ \text{0} \end{pmatrix}$$

define: $\frac{r_1}{r_2} = \alpha$

1) $\ddot{x}_1 - \alpha \ddot{x}_2 = (1 - \alpha) \sin \varphi g$

2) $\ddot{y}_1 - \alpha \ddot{y}_2 = (1 - \alpha) \cos \varphi g$

$\Rightarrow \frac{\ddot{x}_1 - \alpha \ddot{x}_2}{\sin \varphi} = \frac{\ddot{y}_1 - \alpha \ddot{y}_2}{\cos \varphi}$

$\Rightarrow \frac{\ddot{x}_1 - \alpha \ddot{x}_2}{\ddot{y}_1 - \alpha \ddot{y}_2} = \tan \varphi$

imu Scaling:

gyro: $\pm 2000 \text{ } ^\circ/\text{s}$

$$\rightarrow 16.4 \frac{\text{LSB}}{(^{\circ}/\text{s})}$$

$$\Rightarrow \text{scale} = \frac{1}{16.4} \frac{(^{\circ}/\text{s})}{\text{LSB}} \cdot \text{raw} = 0,001064 \frac{\text{rad/s}}{\text{LSB}}$$

accel: $\pm 1g$

$$\rightarrow 8192 \frac{\text{LSB}}{g}$$

$$\Rightarrow \text{scale} = \frac{1}{8192} \frac{g}{\text{LSB}} = \frac{1}{8192} \cdot \frac{9.81 \frac{\text{m}}{\text{s}^2}}{\text{LSB}} = 0,0012$$